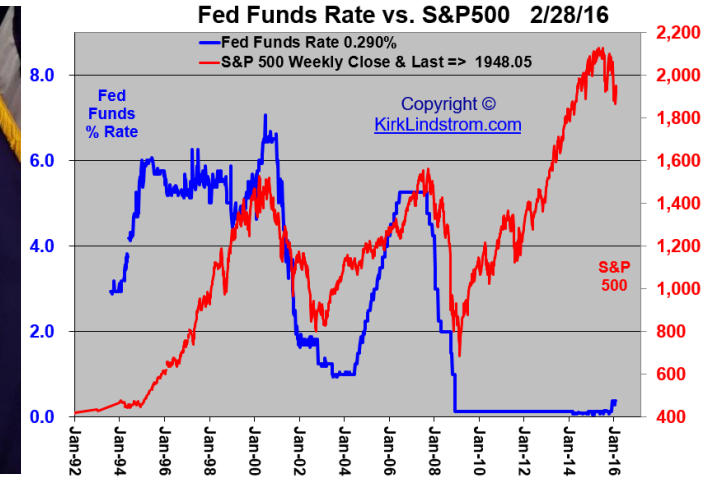


COMP9444: FOMC Sentiment Analysis



NLP Intelligence

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Wing Pui (Nigel) Li	(z5320405)
Zhenhua Liu	(z5433153)
Tong Zhang	(z5542524)

One of the core tasks in Natural Language Processing (NLP) is to extract meaningful stances from complex texts. Our project focuses on extracting policy stances from Federal Open Market Committee (FOMC) meeting notes to classify their positions into categories such as **Neutral, Hawkish, or Dovish** (or {0, 1, 2}). The FOMC's policy stance is a crucial economic indicator that reflects the Federal Reserve's perspective on interest rates, economic growth, and inflation.

Broad array of applications



Correa et al. (2020) found that sentiment in central banks' financial stability reports serves as a **useful predictor of banking crises**, with a 1 percentage point increase in sentiment followed by a 29 percentage point increase in the probability of a crisis



Hilscher et al. (2024) demonstrated that central bank sentiment predicts policy interest rates and the Taylor rule several quarters into the future, indicating a **strong correlation between sentiment and Federal Funds Rate movements**

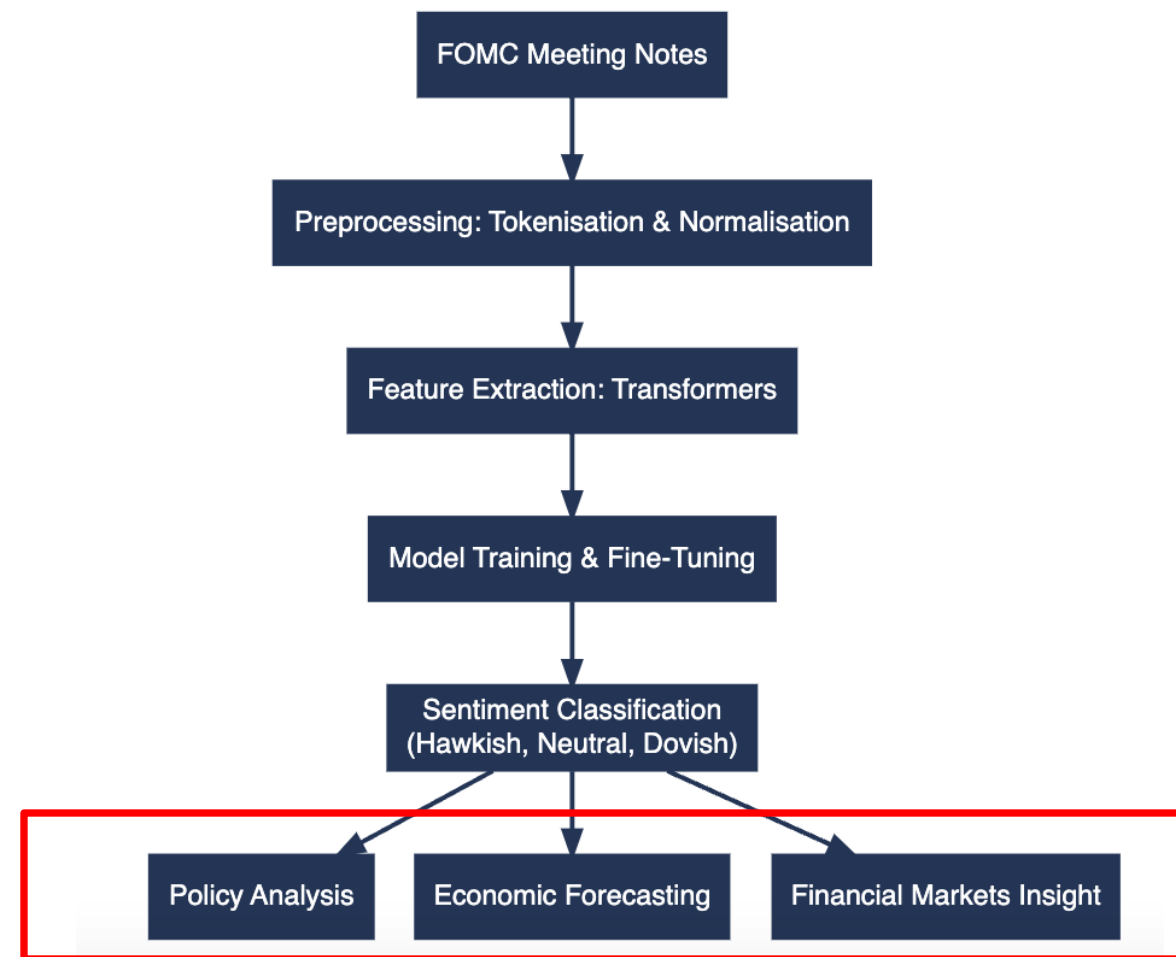


Deng et al. (2024) found that a **one-unit increase in the sentiment score of Federal Reserve communications is associated with an increase of approximately 500 basis points** in the price of the S&P 500 Exchange-Traded Fund, highlighting the significant impact on stock market returns



Correa et al. (2020) developed a financial stability sentiment index using central bank communications from 30 countries, demonstrating that such methodologies can be **applied to emerging markets** to assess financial vulnerabilities and predict crises

Applications of FOMC Sentiment Classification



Problem Statement

The Federal Open Market Committee (FOMC) meeting notes contain critical policy perspectives that shape market expectations and economic forecasting. However, extracting sentiment such as **"Hawkish," "Neutral," or "Dovish"** from these documents requires sophisticated natural language processing techniques. Traditional approaches often fail to fully capture the nuance and complexity inherent in these notes.

There is a need for a robust system to analyse FOMC communications and accurately classify policy stances. **Traditional models** like LSTM often **fail to capture the nuance and complexity** of such texts, leading to unreliable analysis

This project aims to thoroughly evaluate multiple Natural Language Processing (NLP) models for the sentiment classification of FOMC communications and develop an optimised solution leveraging advanced neural architectures and fine-tuning methodologies.

Our objective is to develop a **"best practice" NLP model** that effectively **balances reliability (accuracy, F1 Precision), interpretability, and computational efficiency** for market analysis, in line with guidelines provided by research into best practices for NLP model development (Hirschberg & Manning 2015)

Key Challenges

The intricate and subtle language used in central bank statements often contains **implicit policy stances** that are difficult for generic models to interpret effectively (Nițoi et al., 2023). This requires advanced transformer architectures and sophisticated contextual learning, as explored by Han (2023) and Yang et al. (2022).



Striving for a model that achieves **high accuracy while maintaining computational efficiency** is an ongoing challenge. Studies such as those by Devlin et al. (2019) and Hutter, Kotthoff, & Vanschoren (2019) have highlighted the trade-offs in achieving superior performance without overwhelming computational resources, emphasising the need for parameter optimisation and lightweight fine-tuning



Sources: Hirschberg & Manning (2015); Nițoi, Stanciu & Borlea (2023); Han (2023); Devlin et al. (2019); Hutter, Kotthoff & Vanschoren (2019)

Literature Review: Overall

The literature review is presented through a chronological exploration of key advancements in financial sentiment analysis, highlighting the progression from early dictionary-based approaches to modern transformer-based and ensembling methods, showcasing the evolution of NLP techniques in capturing financial text nuance

Timeline of Literature in Financial Sentiment Analysis



Literature Review

Loughran and McDonald (2011) developed a dictionary-based sentiment analysis for financial texts, which effectively addressed domain-specific sentiment but lacked contextual understanding and flexibility.

Timeline of Literature in Financial Sentiment Analysis



Did: Created dictionary-based financial sentiment analysis.

Limitation: Lacked context-awareness; struggled with nuances.

Literature Review

Greff et al. (2015) introduces LSTMs to capture sequential dependencies in financial text, significantly improving upon traditional models, though computational complexity and long training times were key limitations.

Timeline of Literature in Financial Sentiment Analysis



Did: Utilised LSTMs to effectively model sequential dependencies in financial text.

Limitation: High computational complexity and long training times due to LSTM architecture.

Literature Review

Transformer-based models, like BERT and its variants, have revolutionized financial sentiment analysis by significantly enhancing contextual understanding and capturing language nuances, but face challenges with computational demands and specialized domain adaptation

Timeline of Literature in Financial Sentiment Analysis



Did: Adapted transformer architectures, with better comprehension & overall accuracy.

Limitation: Demanding in computational power, making it resource-heavy for financial analysis.

For example, for 10 combinations of hyperparam searching, it took 25 hours to run on a 3080RTX (for deberta)!

Literature Review

Etelis et al. (2023) demonstrated that ensemble stacking, incorporating diverse NLP model types, improves sentiment analysis performance beyond individual transformer models.

Timeline of Literature in Financial Sentiment Analysis



Did: Used a Hierarchical Ensemble Construction (HEC) algorithm with diverse NLP models, achieving a mean accuracy of 90% + across eight sentiment analysis datasets. → Establish benefit of potential stacking

Limitation: Increased complexity and computational demands compared to using a single transformer model, requiring up to 75 base-learners in evaluation.

Literature Review

Etelis et al. (2023) demonstrated that ensemble stacking, incorporating diverse NLP model types, improves sentiment analysis performance beyond individual transformer models.

Timeline of Literature in Financial Sentiment Analysis



Note that the HEC algorithm is more for models with a hierarchical model structure with each model learning an element from the data, and outputs fed as inputs to higher-level models

Did: Used a Hierarchical Ensemble Construction (HEC) algorithm with diverse NLP models, achieving a mean accuracy of 90% + across eight sentiment analysis datasets. → Establish benefit of potential stacking

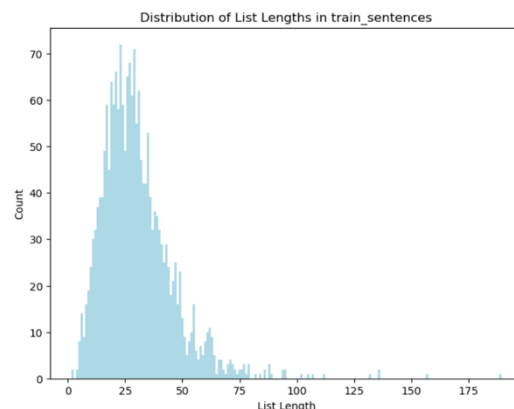
Limitation: Increased complexity and computational demands compared to using a single transformer model, requiring up to 75 base-learners in evaluation.

Dataset & Data Exploration

Dataset from the Financial Services Innovation Lab of Georgia Tech, specifically the FOMC dataset titled 'Trillion Dollar Words: A New Financial Dataset, Task & Market Analysis,' presented at ACL 2023. The team decided to use the "78516" dataset split due to its balance between representativeness and computational feasibility.

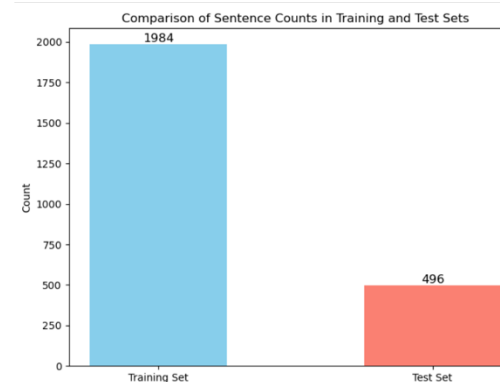
Nature of Data

	A	B	C	D
1	index	sentence	year	label
2	487	Instead,	1998	2
3	422	Inflation	2004	0
4	150	It isn't	2021	2
5	110	At some p	2022	1
6	548	Moreover,	2002	0
7	122	Based on	2007	0
8	282	We have 1	2020	2



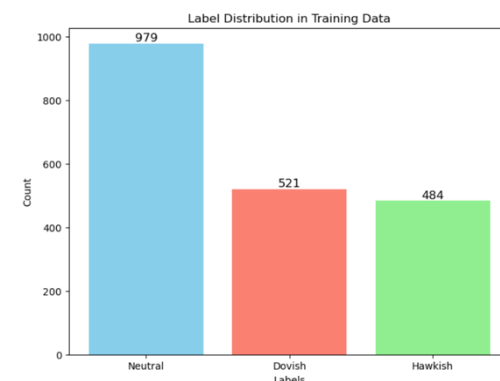
The distribution of list lengths in the training set shows that most lists have fewer than 50 tokens, with a peak around 20-30 tokens. This information is critical for the model to understand the average sentence length, which helps in configuring the appropriate model architecture and parameter settings

Data Split



80/20 Split:

The dataset was split into 80% for training (1,984 samples) and 20% for testing (496 samples), ensuring a standard division for robust training and generalisable model evaluation.



Label Distribution:

Neutral: 979 samples
Dovish: 521 samples
Hawkish: 484 samples

Data Pre-processing

Text Data

Word2Vec (LSTM)

E.g. Our mandate, sorry, is price inflation.

Sentences to word lists Remove case, punctuation

['our', 'mandate', 'sorry', 'is', 'price', 'inflation']

Convert word to vector vector size = 100

'Inflation'

[0.05 0.01 -0.58 0.32 0.68 1.29 -0.27]

Dimension of vector: 100

Convert sentence to list

Set the maximum number of characters to fill the empty field with a vector of zero.

Our mandate, sorry, is price inflation.

...
...
...
...
...

0 [[0.07637715, 0.00684211, -0.3023...
1 [[0.04294358, -0.0225435, -0.1720...
2 [[0.07670072, -0.027039701, -0.38...
3 [[0.018912664, -0.010538941, -0.14...
4 [[0.07670072, -0.027039701,, -0.38...

Tokenization (BERT)

E.g. Our mandate, sorry, is price inflation.

Pretrained tokenizer

Input ID

[101, 2256, 11405, 1010, 3374, 1010, 2003, 3976, 14200, 1012, 102,, 0]

Maximum length of sentence

[CLS]

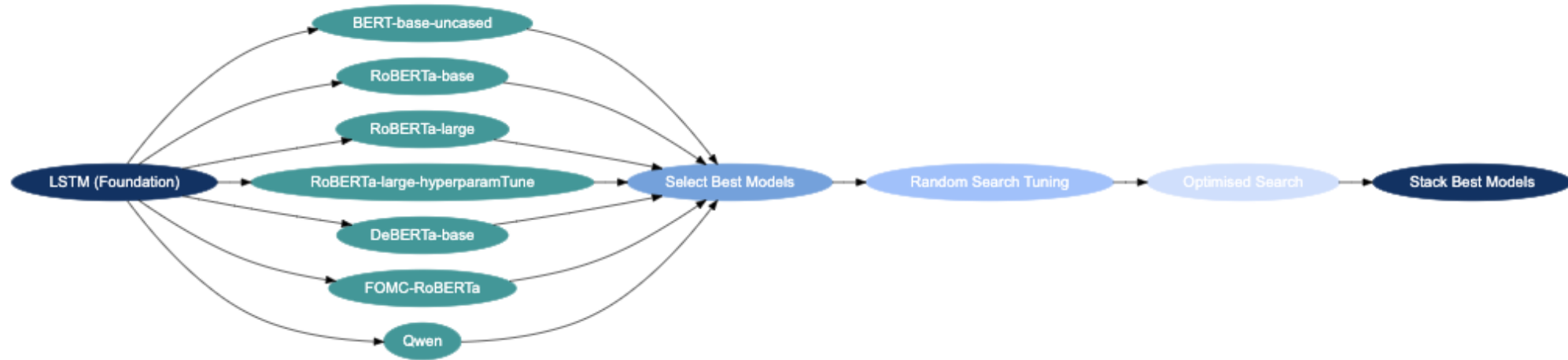
comma

'our'

[SEP]

Padding

Method(s)



Used LSTM (Long Short-Term Memory) model as the **foundational** model to establish a baseline for sentiment analysis

Chosen due to its proven **effectiveness in handling sequential data**, particularly for natural language processing tasks & bc. Learnt in course

Exploration of various **Transformer-based models** to build upon the baseline

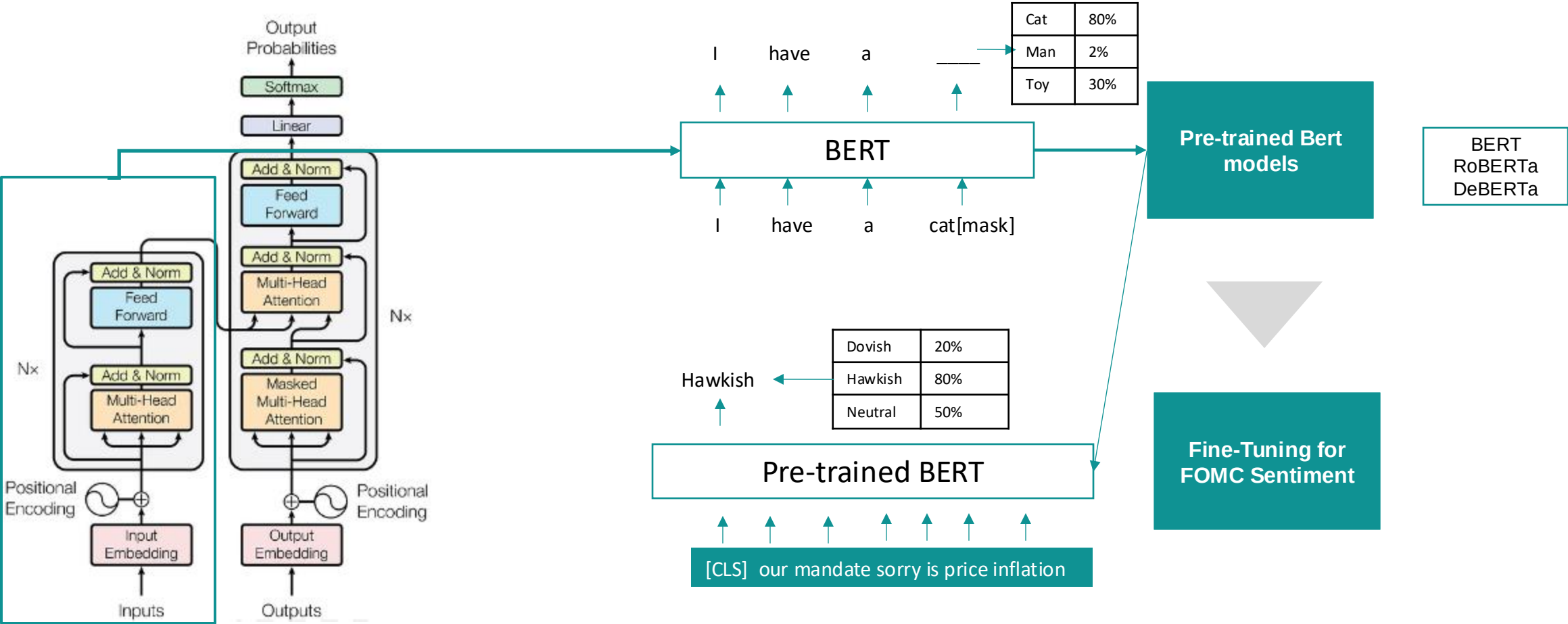
Selection of best models by metrics of {F1, Precision, Accuracy}

1. Random Search Tuning

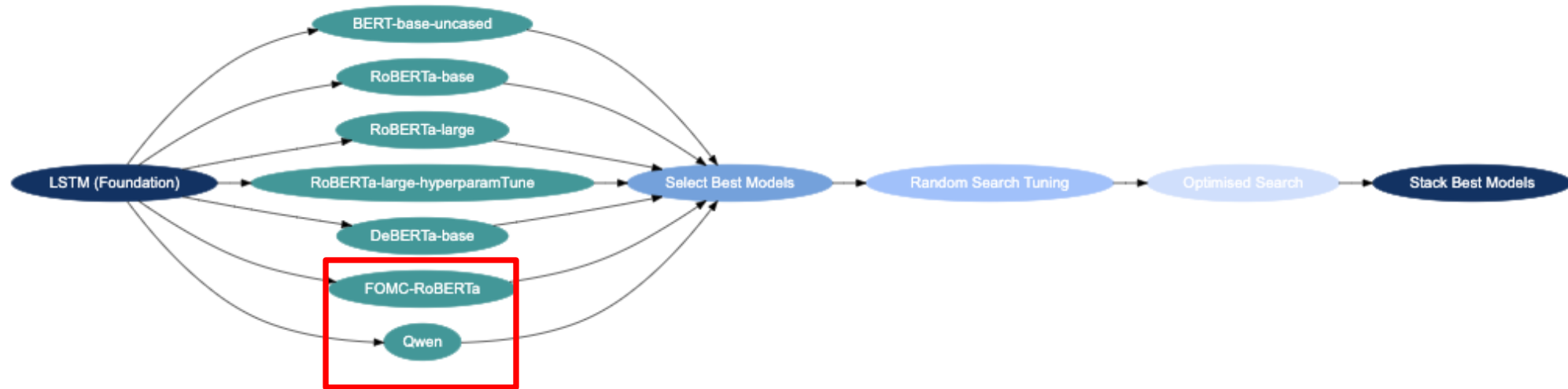
2. Optimised Search

Stacking Phase

Pre-trained BERTs



Method(s)



Used LSTM (Long Short-Term Memory) model as the **foundational** model to establish a baseline for sentiment analysis

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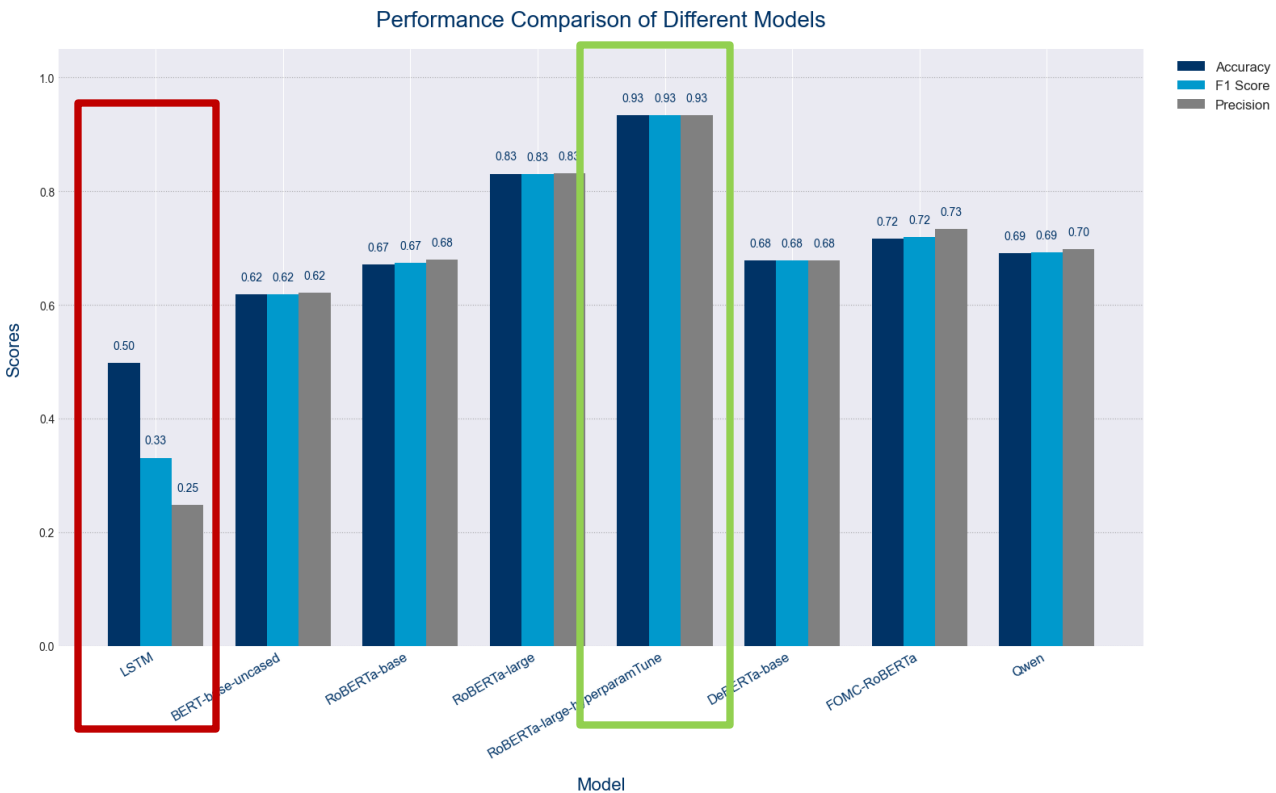
2. Optimised Search

Stacking Phase

Explainability of Performance

Model	Architecture	Features
LSTM	Recurrent Neural Network	Handles sequential data but limited in capturing complex, long-range dependencies. Struggled with long-term dependencies and nuances information.
BERT-base-uncased	Transformer Encoder	Pre-trained model with bidirectional attention; useful for understanding context but limited in comparison to larger models. Struggled evidently, as it had fewer parameters and limited feature extraction capabilities
RoBERTa-base	Transformer Encoder	Improved training over BERT; optimized with larger batch sizes and dynamic masking.
RoBERTa-large	Transformer Encoder	Larger version of RoBERTa-base, he added layers and complexity allow RoBERTa-large to extract deeper features from the input text, giving it an edge in classification tasks where nuanced understanding is essential.
RoBERTa-large-hyperparamTune	Transformer Encoder	Fine-tuned version of RoBERTa-large with optimised hyperparameters, resulting in significantly enhanced performance.
DeBERTa-base	Transformer Encoder with Disentangled Attention	Enhanced handling of dependency between words; handles positional and content-based relationships separately.
FOMC-RoBERTa	Transformer Encoder (Domain-Specific)	Fine-tuned on FOMC-related texts, adapted for specific financial data context. Evidently had limited generalisation ability, and was likely overfit to its existing training data.
Qwen	Transformer Encoder	Performance was limited compared to larger, optimised models due to fewer parameters and general-purpose training.

Performance Comparison of Models



Results – Local Interpretable Model-agnostic Explanations

The fine-tuned RoBERTa-large outperforms LSTM due to its advanced contextual understanding and precision in capturing financial language nuances, highlighting key differences in prediction accuracy and interpretability

Explainability of Performance



RoBERTa-large-hyperparamTune shows high confidence in its classification (100% neutral), whereas LSTM exhibits uncertainty by spreading probabilities across different labels (neutral, dovish, hawkish).



The LSTM model highlights overlapping keywords across multiple labels, indicating weaker feature differentiation. In contrast, RoBERTa-large-hyperparamTune highlights specific words with minimal overlap, suggesting better feature extraction and contextual understanding.



RoBERTa-large-hyperparamTune demonstrates stronger contextual understanding, leading to more decisive classifications, whereas LSTM relies more on individual word-level importance, resulting in less precise predictions

Prediction probabilities

Dovish	0.37
Hawkish	0.36
Neutral	0.27

NOT Dovish

Dovish

NOT Hawkish

Hawkish

NOT Neutral

Neutral

diminish
the
caps
augment
rate
on
ended
forward
curve
balance

diminish
the
augment
caps
ended
forward
on
rate
curve
credibility

diminish
the
caps
augment
rate
on
ended
forward
curve
balance

Text with highlighted words

Setting the horizon on the interest rate caps to reinforce forward guidance on the policy rate would augment the credibility of the yield curve caps and thereby diminish concerns about an open-ended balance sheet commitment.

Prediction probabilities

Dovish	0.00
Hawkish	0.01
Neutral	0.99

NOT Neutral

Neutral

NOT Hawkish

Hawkish

NOT Dovish

Dovish

Setting
thereby
horizon
concerns
the
balance
an
and
guidance
diminish

caps
open
Setting
horizon
ended
credibility
and
thereby
the
reinforce

Setting
concerns
thereby
horizon
ended
an
balance
the
diminish
and

Text with highlighted words

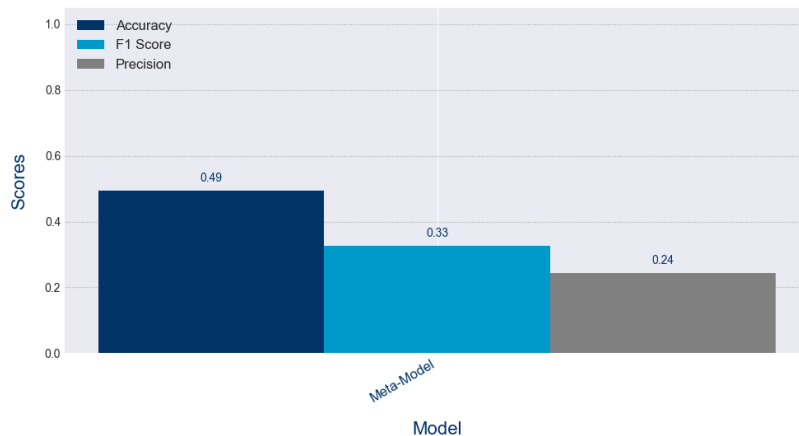
Setting the horizon on the interest rate caps to reinforce forward guidance on the policy rate would augment the credibility of the yield curve caps and thereby diminish concerns about an open-ended balance sheet commitment.

Results – Stacking Result

Etelis et al. (2023) demonstrated that ensemble stacking, incorporating diverse NLP model types, improves sentiment analysis performance beyond individual transformer models. Results did not appear to promising...

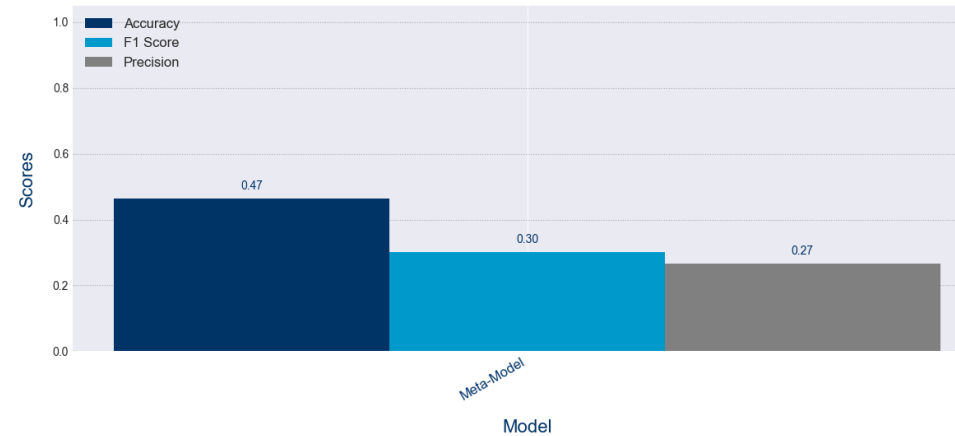
Stacking of FOMC-ROBERTa & ROBERTa-large-hyperparamTune

Performance of Meta-Model (FOMC-RoBERTa + RoBERTa-large-hyperparamTune)



Stacking of ROBERTa-large-hyperparamTune & DeBERTa

Performance of Meta-Model (RoBERTa-large-hyperparamTune + DeBERTa-base)



Why?

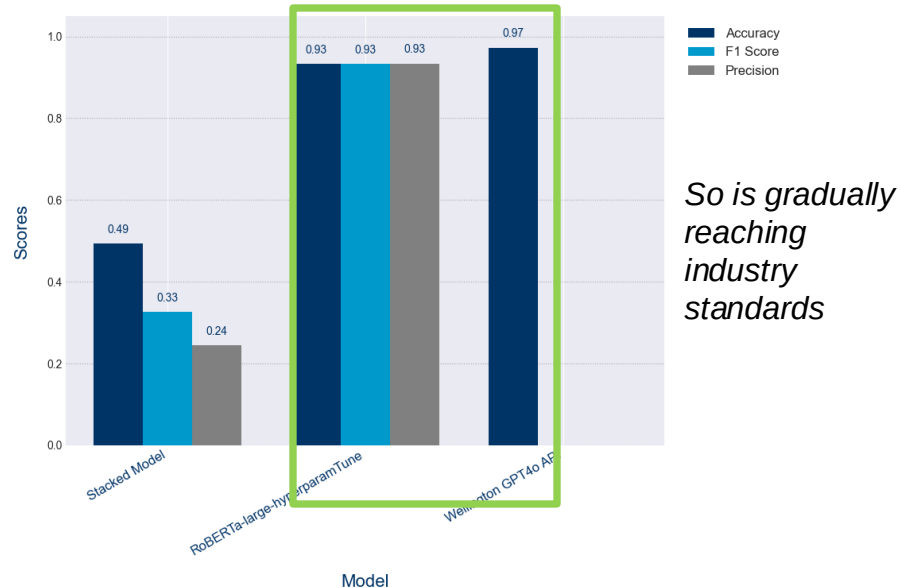
- Advanced transformer techniques with a general large model, combined with disentangled attention & field expertise
- Potentially Complementary
- Strong individual performance, with DeBERTa with the ability to generalise, while RoBERTa's larger pre-training give it the ability to understand large constructs. Likewise, with FOMC-RoBERTa

The stacking of FOMC-RoBERTa with RoBERTa-large, as well as DeBERTa with RoBERTa-large, has not yielded strong results primarily due to **limited complementary strengths between the models**, where overfitting in domain-specific fine-tuned models (like FOMC-RoBERTa) and insufficient differentiation in feature extraction from similar Transformer structures reduce the diversity and generalisation advantages typically sought in stacked models.

A wide range of transformer-based models were evaluated with LSTM as the base model, including BERT, RoBERTa, DeBERTa and others, to assess their ability to classify sentiment in FOMC texts. A gradual iterative process did lead us to eventually reach almost-industry levels of accuracy....

Comparison with Industry

Performance Comparison: Stacked Model vs RoBERTa-Large-HyperparamTune vs Wellington Figure



Source: Wellington Management Quantitative Investment Group (Julia Klevak, Research Scientist)

Key Challenges

Fine Tuning Difficulties

- Varied effectiveness for domain-specific texts (e.g., FOMC).
- Hyperparameter selection was challenging; risk of overfitting (e.g., FOMC-RoBERTa).



Computational Expensiveness

- Training RoBERTa-large (25 hours on 3070 RTX GPU for hyperparam tuning of 10 combinations).
- High resource demand limits scalability.



Performance Trade-offs with Stacking Models

- Limited gains from stacking (RoBERTa-large + FOMC-RoBERTa).
- Increased model variance; high sensitivity to training data.
- Computationally intensive without consistent performance benefits.



Conclusion

Consolidation of Insights: Evaluating Model Efficiency, Optimal Approaches, and Future-Ready Solutions for Sentiment Analysis...



Stacking Performance

- Stacking transformer models did not outperform individual fine-tuned models.
- Individual model (RoBERTa-large-hyperparamTune) showed better consistency.



Best Practice Identified

- Single, well-tuned model (RoBERTa-large-hyperparamTune) proved optimal for this certain case. Although industry trends does appear to currently utilise GPT-4o, which is another transformer architecture
- Ensemble techniques added complexity without consistent gains.



Future Applications

- **Exploring larger and more sophisticated models like GPT-4-tuned versions** or experimenting with transformer models that have demonstrated high success rates across other domains could help surpass the current 93% accuracy threshold
- **Incorporating additional financial and economic datasets**, including sentiment data, textual context, and structured numerical information, might enhance model precision beyond current limitations

A nighttime photograph of the National Congress of Argentina, a large neoclassical building with many lit windows. In the foreground, there are horizontal light trails from traffic, primarily in shades of red and yellow. The word "Appendices" is centered in the lower half of the image.

Appendices

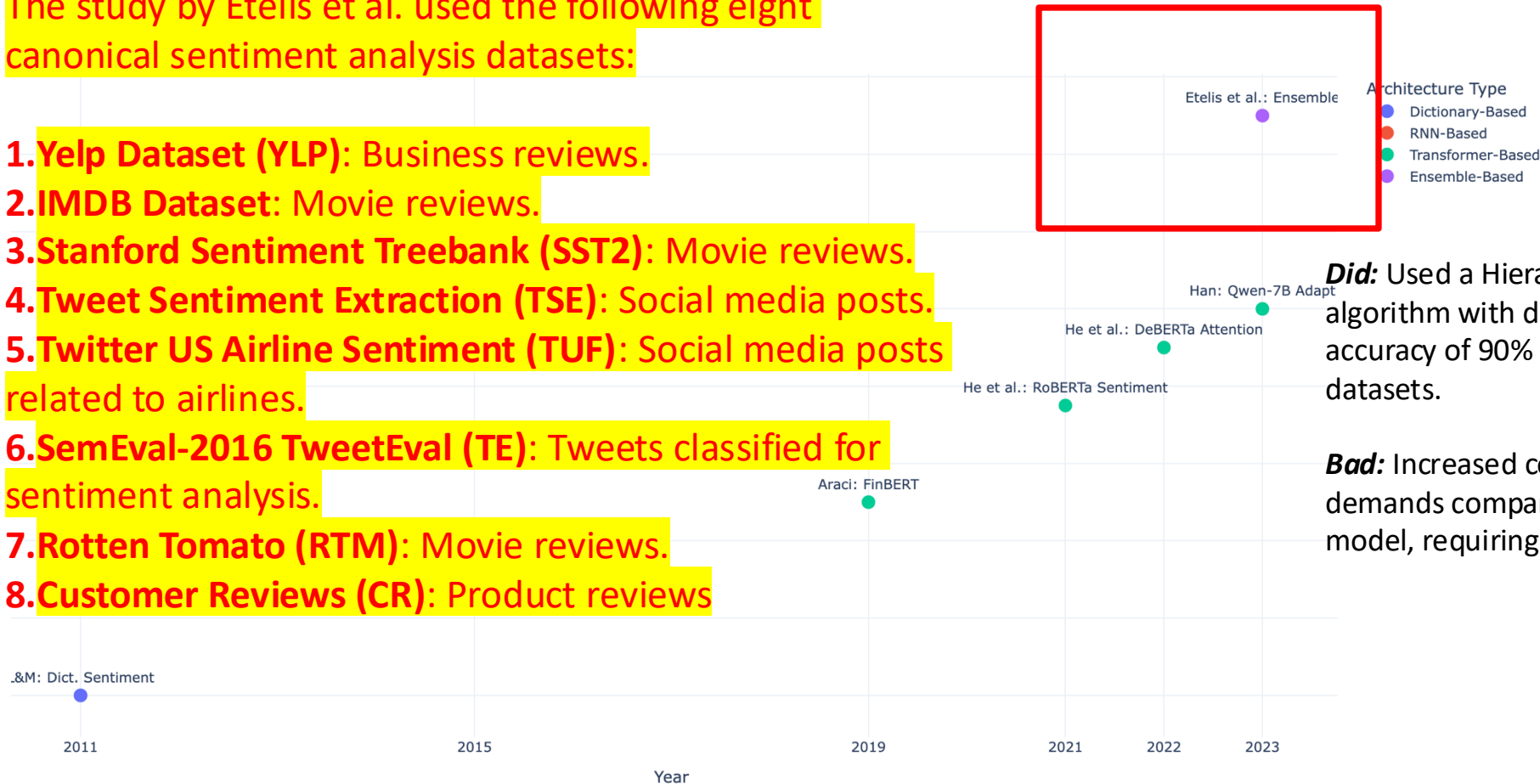
Presentation	Introduction	1	Presentation	Results	Appendix	Appendix Overview	17	
	Set-up			Overview		12	Literature Review – Datasets used	18
	Motivation	2		Results - LIME		13	LSTM Architecture	19
	Problem Statement	3		Results - Stacking		14	RoBERTa Architecture	20
							DeBERTa Architecture	21
	Literature Review			Discussion		15	Qwen Architecture	22
	Overall	4					Training with RTX3080	23
	Literature Review: Dictionaries	5					Hyperparameter Tuning Rationale	24
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Literature Review – Datasets Used

Etelis et al. (2023) demonstrated that ensemble stacking, incorporating diverse NLP model types, improves sentiment analysis performance beyond individual transformer models.

The study by Etelis et al. used the following eight canonical sentiment analysis datasets:

1. **Yelp Dataset (YLP):** Business reviews.
2. **IMDB Dataset:** Movie reviews.
3. **Stanford Sentiment Treebank (SST2):** Movie reviews.
4. **Tweet Sentiment Extraction (TSE):** Social media posts.
5. **Twitter US Airline Sentiment (TUF):** Social media posts related to airlines.
6. **SemEval-2016 TweetEval (TE):** Tweets classified for sentiment analysis.
7. **Rotten Tomato (RTM):** Movie reviews.
8. **Customer Reviews (CR):** Product reviews

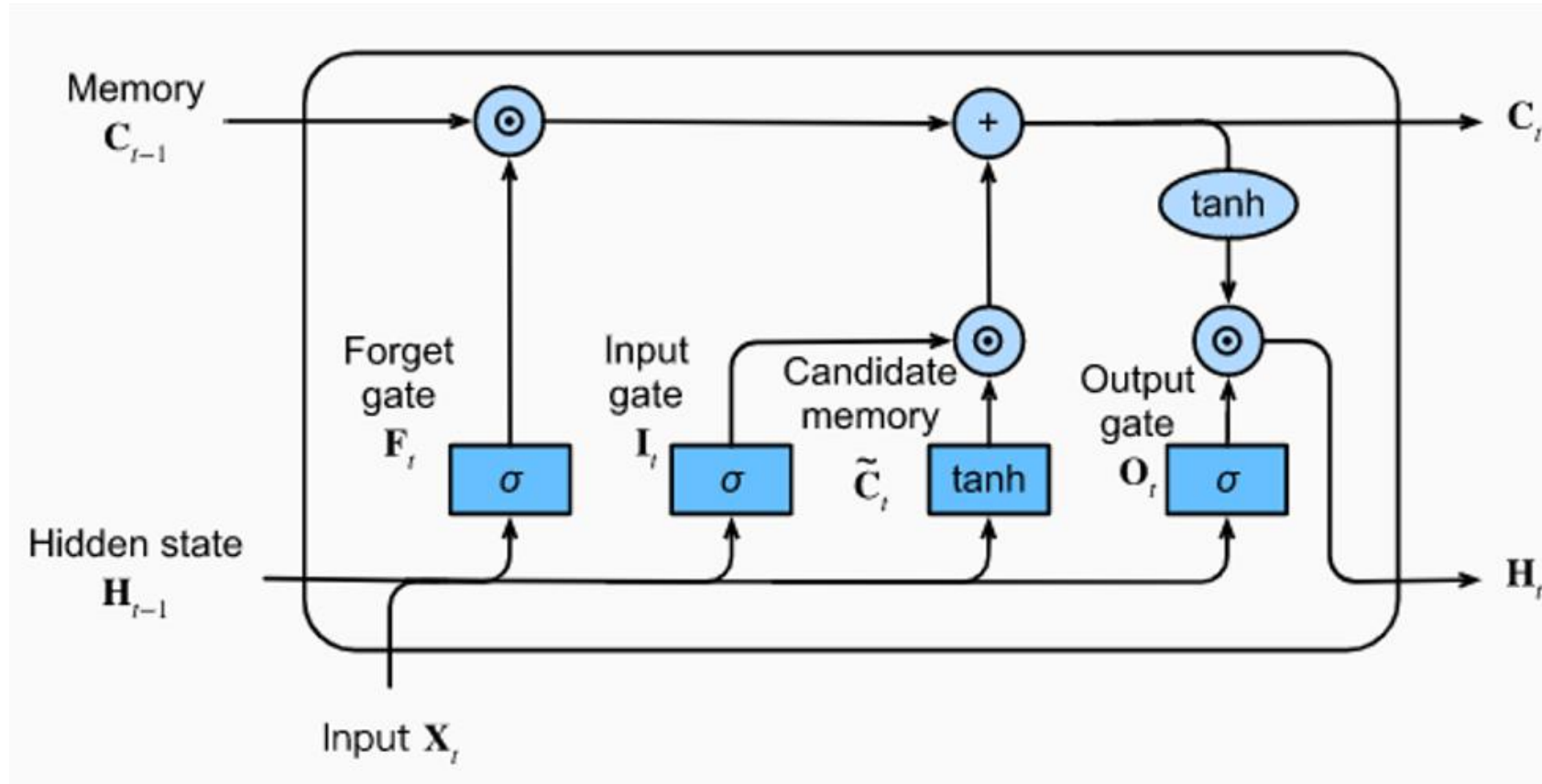


Did: Used a Hierarchical Ensemble Construction (HEC) algorithm with diverse NLP models, achieving a mean accuracy of 90%+ across eight sentiment analysis datasets.

Bad: Increased complexity and computational demands compared to using a single transformer model, requiring up to 75 base-learners in evaluation.

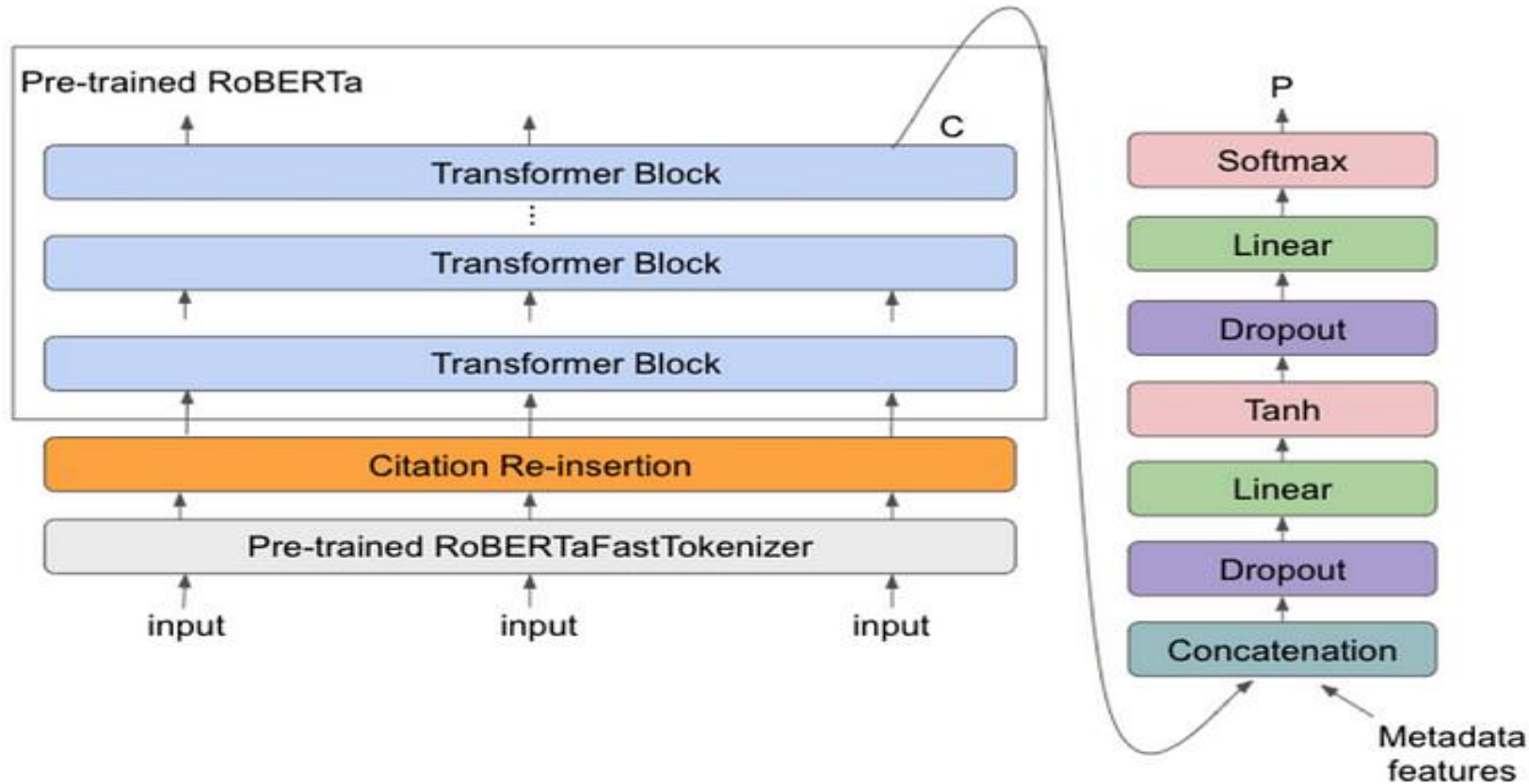
LSTM Architecture

LSTM was utilised as a baseline model for our NLP analysis



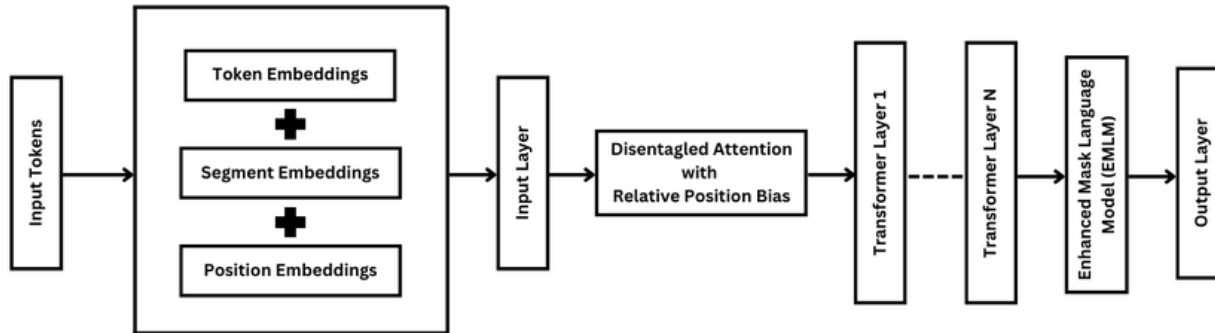
RoBERTa Architecture

RoBERTa proved to be the most optimal model



DeBERTa Architecture

DeBERTa was initially promising, but did appear to have its issues...



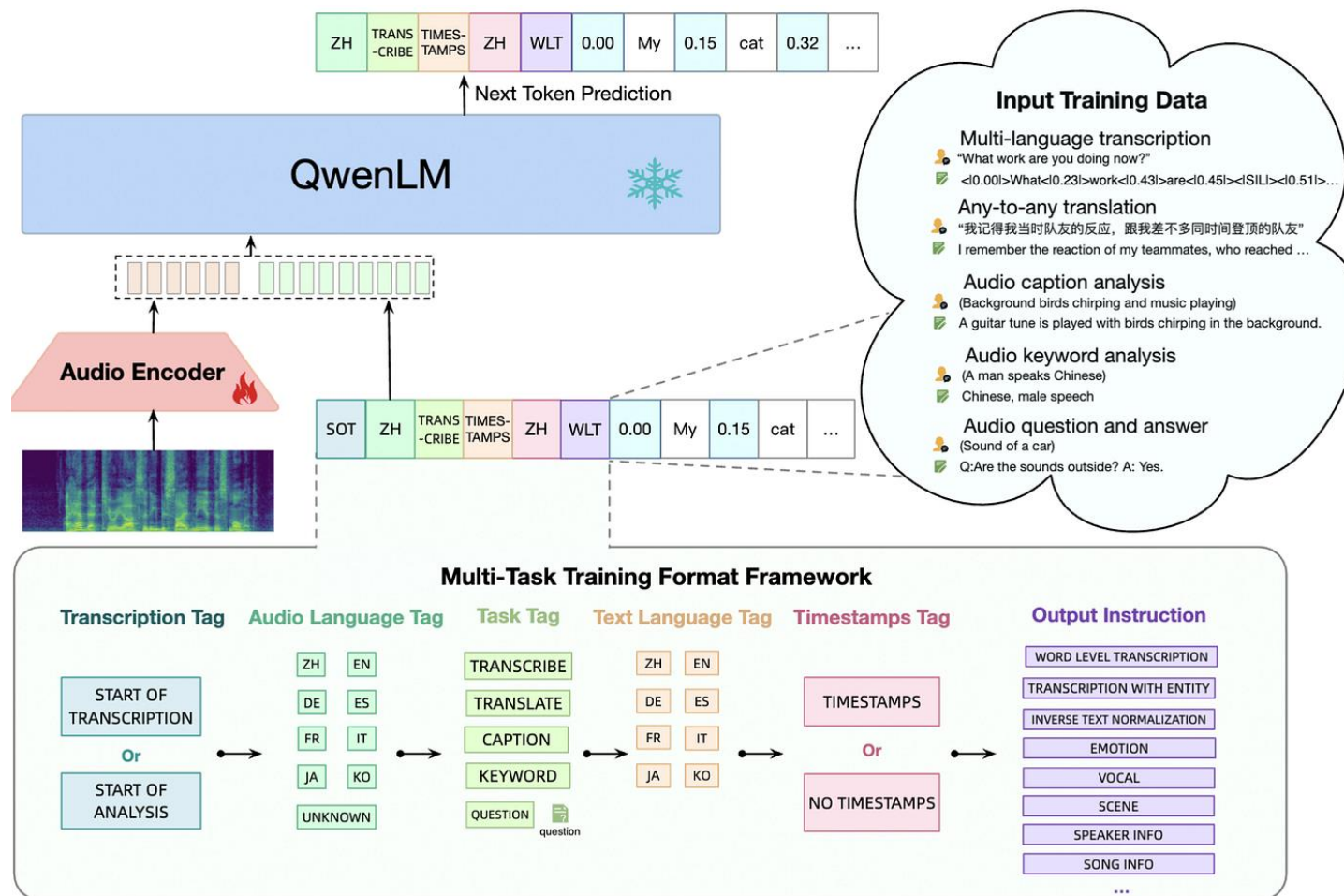
Disentangled Attention: DeBERTa separates (or “disentangles”) the content of words from their positions. It does this by encoding each word's content and its positional information independently, allowing the model to pay attention to both aspects more effectively. This improves the model's ability to comprehend relationships between words in context.

Relative Position Embedding: DeBERTa’s attention mechanism includes relative position embeddings rather than absolute ones, which allows it to capture long-range dependencies more effectively than traditional transformers.

Improved Masked Language Model: DeBERTa enhances the masked language modeling process, learning better contextual representations that help with nuanced understanding, considering {relative position, content} in its embeddings

Qwen Architecture

Qwen was evaluated as one of the potential architectures



Training with RTX 3080 w. CUDA

Demonstrating computational expensiveness

```
import torch
print("GPU is available:", torch.cuda.is_available())
print("GPU Device Count:", torch.cuda.device_count())
print("GPU Device Name:", torch.cuda.get_device_name(0) if torch.cuda.is_available() else "No GPU detected")
device = torch.device("cuda" if torch.cuda.is_available() else "cpu")

torch.cuda.empty_cache()
torch.cuda.set_per_process_memory_fraction(0.99)
```

6]

```
.. GPU is available: True
GPU Device Count: 1
GPU Device Name: NVIDIA GeForce RTX 3080
```

Hyperparameter Tuning Rationale

The process of fine-tuning involved random search, then optimised search with more optimised parameters ...

Hyperparameter	Description	Impact on Model
Learning Rate	Controls how much to change the model in response to the estimated error each time model weights are updated.	A higher learning rate can speed up training but risks overshooting the optimal solution. A lower rate provides stable training but can be very slow.
Batch Size	Number of training examples utilized in one iteration.	Larger batch sizes offer more stable gradients but require more memory. Smaller batches may lead to better generalization but noisier training.
Epochs	Number of times the entire dataset is passed forward and backward through the model.	More epochs allow the model to learn better, but too many can cause overfitting.
Max Length	Maximum number of tokens considered from the input text.	A larger maximum length allows for longer context, which improves understanding but increases computation and memory usage.
Hidden Dropout	Dropout applied to fully connected layers during training to prevent overfitting.	Higher dropout values reduce overfitting risk but can slow down convergence. Too low a value might lead to overfitting.
Attention Dropout	Dropout applied to the attention mechanism to regularize and prevent overfitting.	Helps the model to not rely too heavily on particular attention weights, promoting robustness. However, too much dropout can hamper learning.

Fine Tuning Process for RoBERTa-Large

The process of fine-tuning involved random search, then optimised search with more optimised parameters ...

Random Search

Trial	Learning Rate	Batch Size	Epochs	Max Length	Hidden Dropout	Attention Dropout	Train Accuracy	Test Accuracy	Train F1 Score	Test F1 Score
1	0.00001	16	10	50	0.1	0.1	0.997479839	0.699596774	0.997481661	0.709971699
2	0.00001	16	10	75	0.1	0.1	0.991935484	0.703629032	0.991946733	0.725743208
3	0.00002	16	10	50	0.1	0.1	0.993951613	0.701612903	0.993949791	0.721192025
4	0.00002	16	10	75	0.1	0.1	0.996975806	0.697580645	0.996976208	0.724541632
5	0.00003	16	10	50	0.1	0.1	0.996975806	0.70766129	0.996978383	0.708563043
6	0.00003	16	10	75	0.1	0.1	0.996975806	0.693548387	0.996977713	0.70122742
7	0.00001	16	10	50	0.3	0.1	0.993447581	0.693548387	0.993445341	0.709621421
8	0.00001	16	10	75	0.3	0.1	0.996471774	0.717741935	0.996471601	0.720427922
9	0.00002	16	10	50	0.3	0.1	0.994959677	0.675403226	0.994959899	0.683569682
10	0.00002	16	10	75	0.3	0.1	0.997479839	0.721774194	0.997481661	0.720499973
11	0.00003	16	10	50	0.3	0.1	0.992943548	0.66733871	0.992946063	0.693947225
12	0.00003	16	10	75	0.3	0.1	0.982358871	0.697580645	0.982331028	0.705469479
13	0.00001	16	15	50	0.1	0.1	0.996471774	0.697580645	0.996469958	0.713963077
14	0.00001	16	15	75	0.1	0.1	0.997479839	0.695564516	0.997480755	0.709306904
15	0.00002	16	15	50	0.1	0.1	0.987903226	0.695564516	0.987921546	0.715365944
16	0.00002	16	15	75	0.1	0.1	0.996975806	0.701612903	0.996976571	0.710024361
17	0.00003	16	15	50	0.1	0.1	0.98891129	0.685483871	0.988961532	0.712546618
18	0.00003	16	15	75	0.1	0.1	0.997479839	0.675403226	0.997480693	0.694066736
19	0.00001	16	15	50	0.3	0.1	0.996975806	0.70766129	0.996978197	0.712880709
20	0.00001	16	15	75	0.3	0.1	0.995967742	0.695564516	0.995969855	0.704778857
21	0.00002	16	15	50	0.3	0.1	0.997479839	0.717741935	0.997478847	0.721955313
22	0.00002	16	15	75	0.3	0.1	0.501008065	0.463709677	0.334454007	0.311728884
23	0.00003	16	15	50	0.3	0.1	0.996975806	0.689516129	0.996973986	0.704353893
24	0.00003	16	15	75	0.3	0.1	0.968245968	0.66733871	0.967989039	0.702222273
25	0.00001	16	10	50	0.1	0.3	0.99546371	0.699596774	0.995462115	0.707102133
26	0.00001	16	10	75	0.1	0.3	0.99546371	0.691532258	0.995464162	0.700391886
27	0.00002	16	10	50	0.1	0.3	0.996975806	0.701612903	0.996978647	0.704910236
28	0.00002	16	10	75	0.1	0.3	0.996975806	0.72983871	0.996976998	0.732300639
29	0.00003	16	10	50	0.1	0.3	0.992439516	0.695564516	0.992447592	0.712676856
30	0.00003	16	10	75	0.1	0.3	0.997479839	0.713709677	0.997478847	0.722853904
31	0.00001	16	10	50	0.3	0.3	0.501008065	0.463709677	0.334454007	0.342103438
32	0.00001	16	10	75	0.3	0.3	0.993447581	0.719758065	0.993441503	0.724020517
33	0.00002	16	10	50	0.3	0.3	0.997479839	0.721774194	0.997481661	0.721724738
34	0.00002	16	10	75	0.3	0.3	0.994959677	0.697580645	0.994957114	0.713229674
35	0.00003	16	10	50	0.3	0.3	0.501008065	0.463709677	0.334454007	0.602764763
36	0.00003	16	10	75	0.3	0.3	0.995967742	0.691532258	0.995969207	0.698922677
37	0.00001	16	15	50	0.1	0.3	0.997479839	0.695564516	0.997478913	0.704080766
38	0.00001	16	15	75	0.1	0.3	0.996975806	0.699596774	0.996976805	0.717319432
39	0.00002	16	15	50	0.1	0.3	0.990927419	0.709677419	0.99091089	0.737857247
40	0.00002	16	15	75	0.1	0.3	0.997479839	0.697580645	0.997481661	0.714125863
41	0.00003	16	15	50	0.1	0.3	0.995967742	0.691532258	0.995966852	0.713327285
42	0.00003	16	15	75	0.1	0.3	0.995967742	0.715725806	0.995968769	0.726517103
43	0.00001	16	15	50	0.3	0.3	0.995967742	0.675403226	0.995969674	0.691591328
44	0.00001	16	15	75	0.3	0.3	0.996975806	0.737903226	0.996976208	0.743978736
45	0.00002	16	15	50	0.3	0.3	0.995967742	0.677419355	0.995966403	0.714873911
46	0.00002	16	15	75	0.3	0.3	0.996471774	0.693548387	0.996475524	0.733600156
47	0.00003	16	15	50	0.3	0.3	0.983366935	0.66733871	0.983324658	0.711026424
48	0.00003	16	15	75	0.3	0.3	0.501008065	0.463709677	0.334454007	0.293810539

Bayesian Optimisation (structured like a greedy-epsilon policy)

- Builds an iteratively probabilistic model that maps hyperparameters, after the random search we set ranges
- More efficient as it explored promising hyperparameter values
- Less computationally expensive than grid search

Note, this random search process took 25 hours...

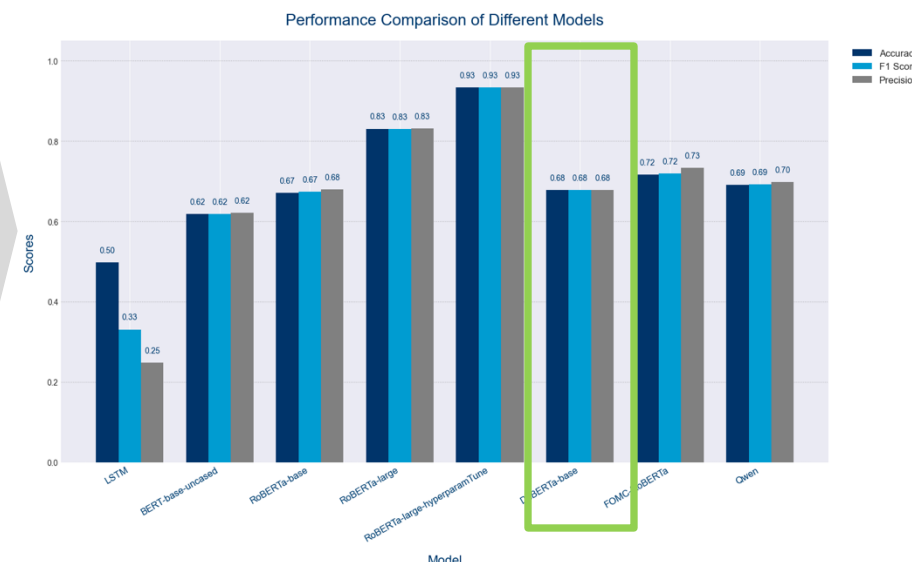
Fine Tuning Process for DeBERTa-Large

The process of fine-tuning involved random search, then optimised search with more optimised parameters ...

Random Search

Trial	Learning Rate	Batch Size	Epochs	Max Length	Hidden Dropout	Attention Dropout	Test Accuracy	Test F1 Score
1	1.00E-05	8	3	128	0.1	0.1	0.514112903	0.475532808
2	3.00E-05	8	5	128	0.2	0.2	0.737903226	0.738384111
3	2.00E-05	8	5	256	0.2	0.1	0.747983871	0.748117281
4	1.00E-05	8	3	128	0.2	0.1	0.752016129	0.750811226
5	1.00E-05	8	5	256	0.2	0.1	0.737903226	0.737606474
6	1.00E-05	16	3	256	0.2	0.1	0.731854839	0.732716836
7	1.00E-05	16	3	128	0.1	0.2	0.719758065	0.718901159
8	3.00E-05	8	3	256	0.1	0.2	0.745967742	0.745317183
9	1.00E-05	8	3	256	0.1	0.1	0.741935484	0.741859184
10	2.00E-05	8	3	256	0.1	0.1	0.733870968	0.734802007

Post Bayesian Optimisation



Failed to generalise to other test sets

Note, this random search process took 26 hours...

Stacking Considerations

The process of stacking was initially proposed to help enhance and further the current outcomes...

Stacking Consideration	Description	Supporting Study	Application to Model Combinations
Model Complementarity	Combining models that capture different linguistic features enhances performance.	"Natural Language Inference with Transformer Ensembles and Explainability Techniques" ArXiv	RoBERTa-large and DeBERTa have distinct architectures; stacking them can harness their unique strengths. Similarly, FOMC-RoBERTa (domain-specific) combined with RoBERTa-large (general-purpose) balances domain adaptation with general language understanding.
Stacking Architecture	The meta-learner should effectively integrate base model outputs. Logistic regression or shallow neural networks are commonly used.	"Optimizing Multi-Class Text Classification: A Diverse Stacking Ensemble Framework" ArXiv	Utilizing a logistic regression meta-learner can effectively combine outputs from RoBERTa-large and DeBERTa, as well as FOMC-RoBERTa and RoBERTa-large, ensuring balanced integration of their predictions.
Avoiding Overfitting	Ensuring base models generalize well and their errors are uncorrelated is crucial. Cross-validation during stacking helps mitigate overfitting.	"Improving Academic Skills Assessment with NLP and Ensemble Learning" ArXiv	Implementing cross-validation in stacking RoBERTa-large with DeBERTa, and FOMC-RoBERTa with RoBERTa-large, can reduce overfitting risks by ensuring the meta-learner is robust.

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